



MXCuBE at P11 - status and challenges

P11 beamline, DESY Hamburg

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MXCuBE/ISPyB Meeting Hamburg
20.05.2025

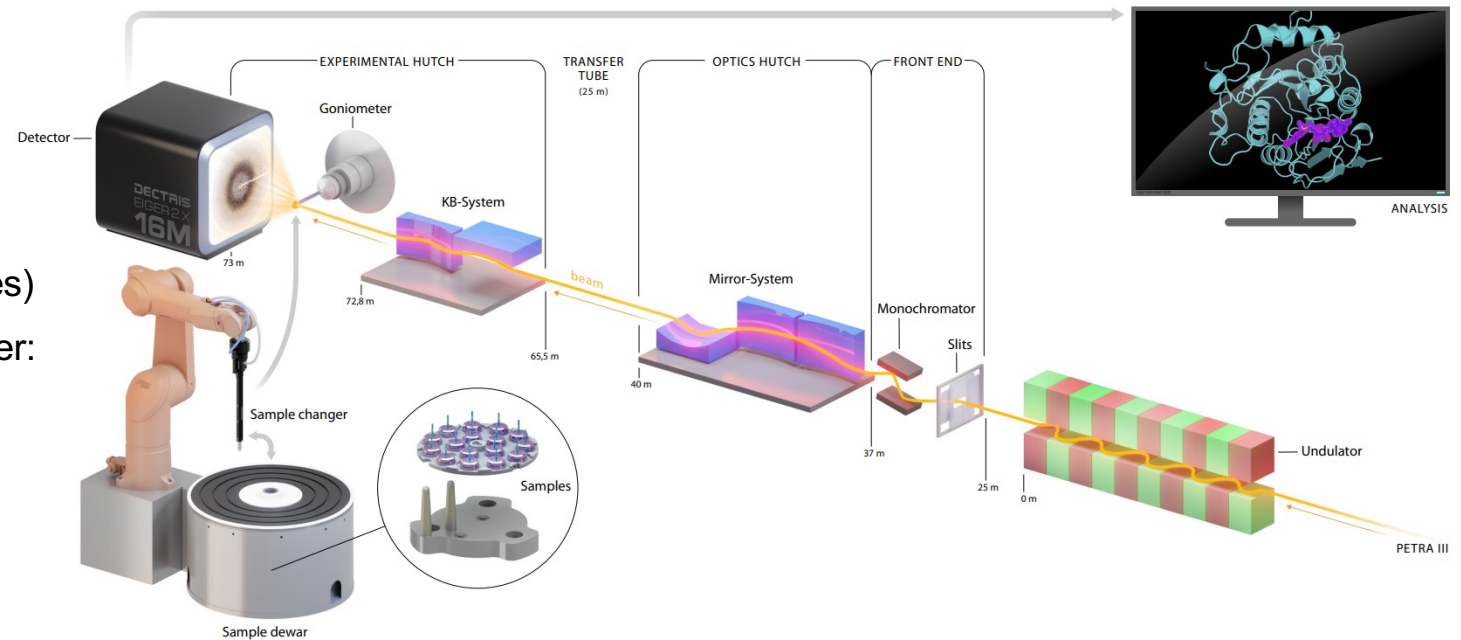
HELMHOLTZ



The High-throughput Macromolecular Crystallography Beamline P11

PETRA III

- In user operation since 2013
- Broad energy range: 5.5 - 28 keV
- High-speed sample changer with capacity (23 unipucks = 368 samples)
- High precision single axis goniometer: 0.0001° at $120^\circ/\text{s}$
- Eiger2 X 16M
- XRF for experimental phasing (SAD/MAD) and element analysis



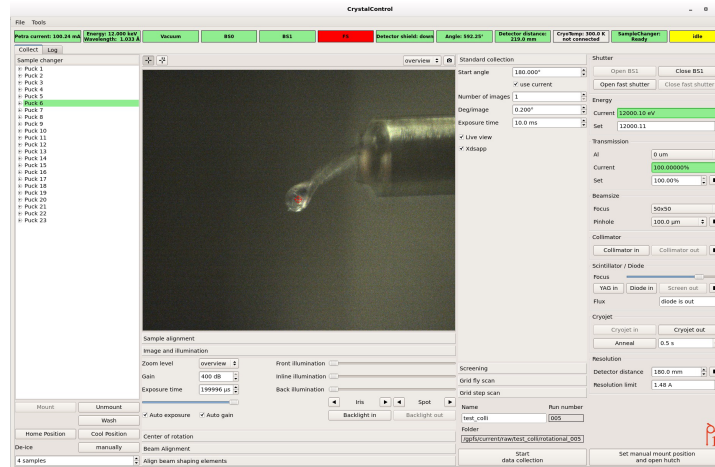
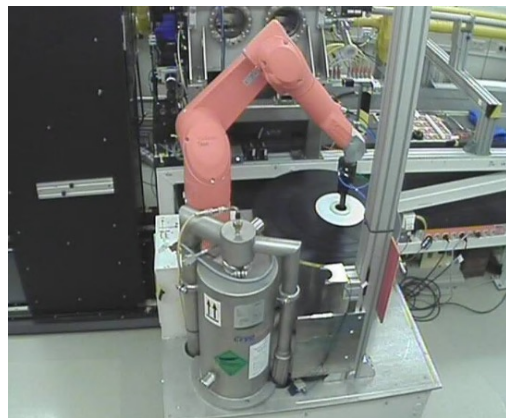
MXCuBE predecessor - CrystalControl



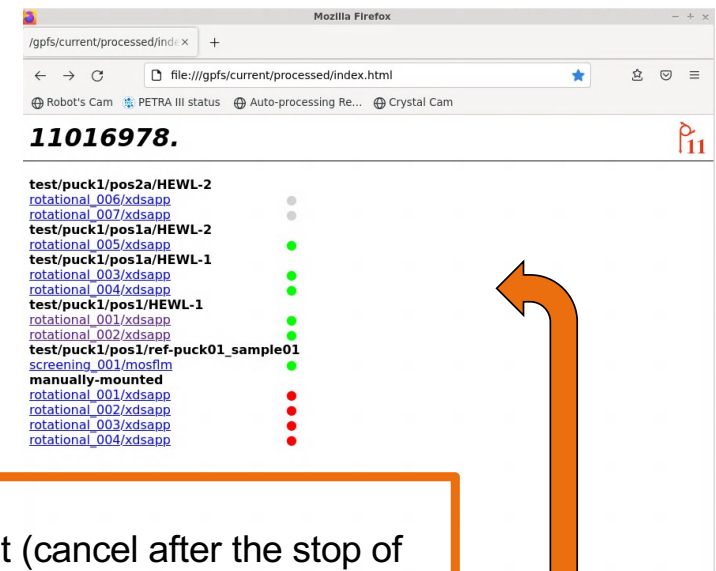
Hardware controls

Tango servers

- Maintained locally
- Maintained centrally (FS-EC)



Results presentation



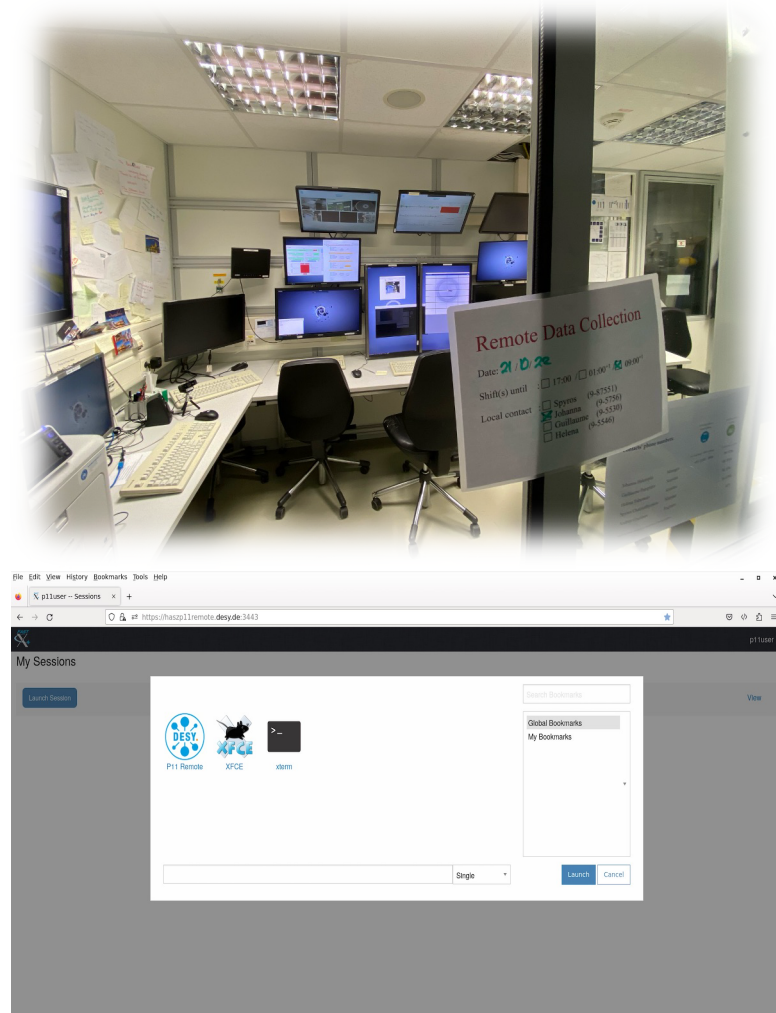
Maxwell Cluster

- "Processing" user assignment (cancel after the stop of the beamtime)
- Resource allocation (nodes)
- xdsapp
- autoPROC

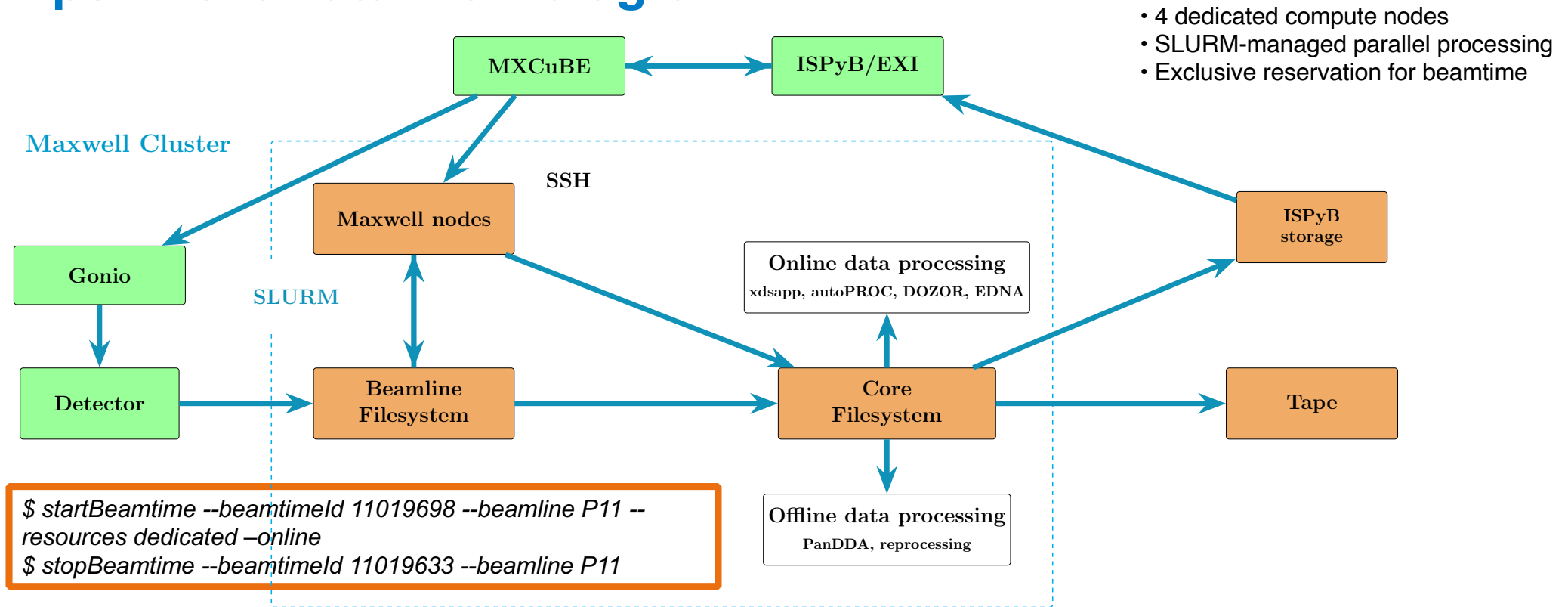
Experiment Workflow in MXCuBE at P11

Remote mode for most of the users

- **FastX**
 - Commercial software with available licenses
 - Web-Based – no additional software
 - Shareable sessions
 - Dedicated host
- **Tailored X Session**
 - "Kiosk mode" KDE environment
 - MXCuBE GUI for managing experiments
 - Browser for accessing results, cameras, and wikis
 - Shareable link via user portal
- **No Terminal**



Experiment Data Flow diagram



GPFS Core Storage

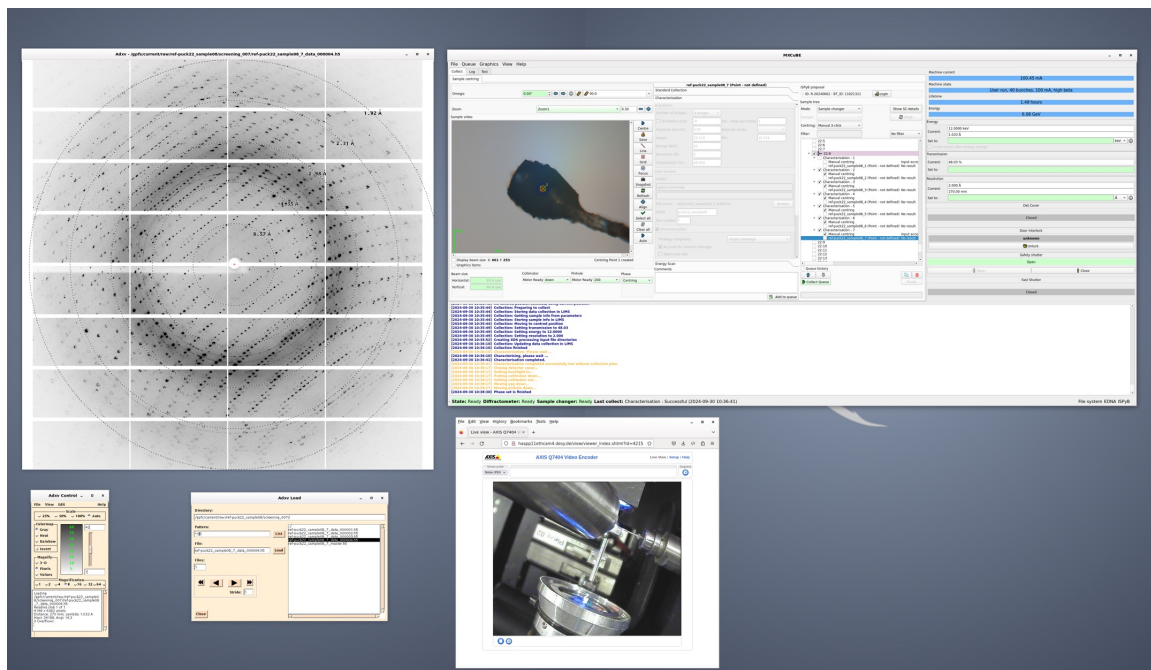
- 180 days retention (2 yrs in-house)
- Frequent file snapshots (protection)

Long-term Archiving & Export

- Copied to dCache and 2 tape copies
- User access via portals, Globus
- Data restore on request

Current status

MXCuBE - Macromolecular Xtallography Customized Beamline Environment



- **Upgraded** to the latest development branch from November 2024, collaboration with B. Rey (ESRF)
- Sample name synchronization and basic results display in **ISPyB**
- **Data collection**
- **Online data analysis** with custom scripts
- Implemented basic functionality of **EDNA Characterization** (LTP G. Bourenkov, EMBL Hamburg)
- Exploring AI-based **optical centring** (LTP T. White, collaboration with M. Savko, Soleil)
- Production and development branches (P11)
- Vanilla Python compiled from source + venv in a shared folder.

<https://github.com/mxcube>

Sample Centring: Lessons from Murko Integration

Collaboration with M.Savko (Soleil) and T.White (FS-SC)

Status:

- Installed and tested locally and on the dedicated node
- Update of the model with P11-specific images
- Initial timing checks
- Local installation is enough for per-image usage

Challenges:

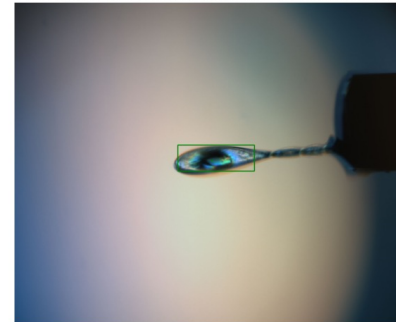
- A larger field of view makes Kapton loop recognition harder.
- The algorithm often confuses the lower opening with the actual loop.
- Works better with a narrower field of view.
- Speed is limited by motor movement

Implementation Goals

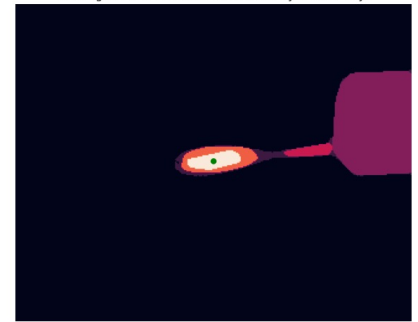
- Implement raster scanning
- Incorporate bounding box coordinates for the area scan and analysis

<https://github.com/MartinSavko/murko>

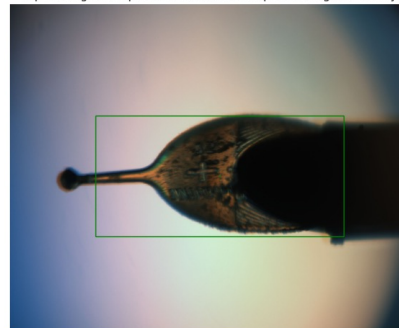
input image with predicted click and loop bounding box (if any)



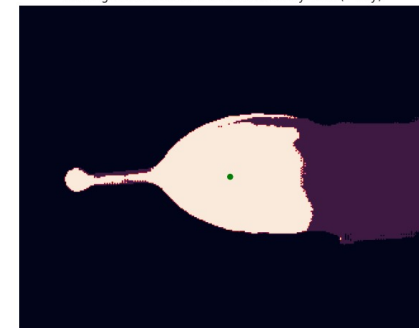
raw segmentation result with most likely click (if any)



input image with predicted click and loop bounding box (if any)

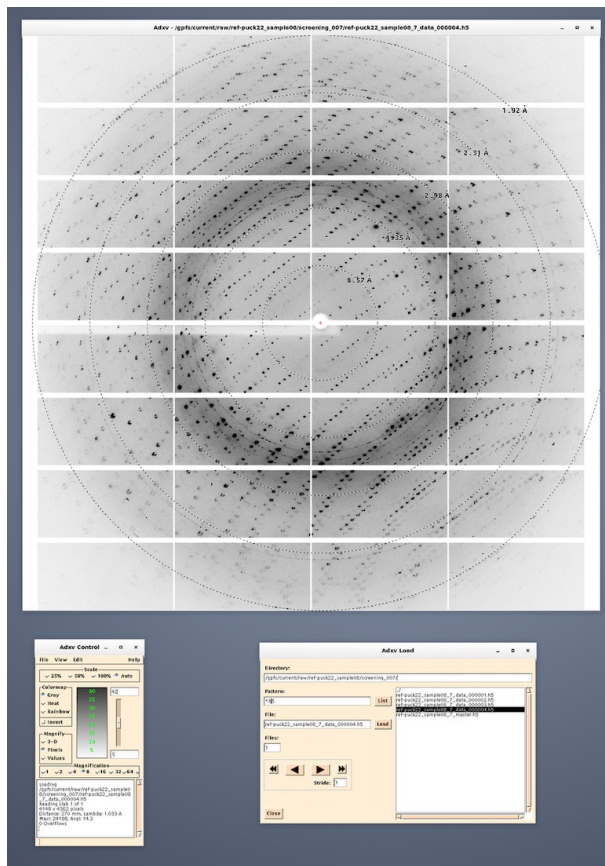


raw segmentation result with most likely click (if any)



Real-time Data Display Integration

Yet Another MONitoring interface for the Eiger + ADXV as “live view”



Live Visualization of Diffraction Data

- Yamone acts as a bridge between the Eiger 16M detector and ADXV visualization software.
- Provides instant display of diffraction images as data is collected.

Technical Implementation

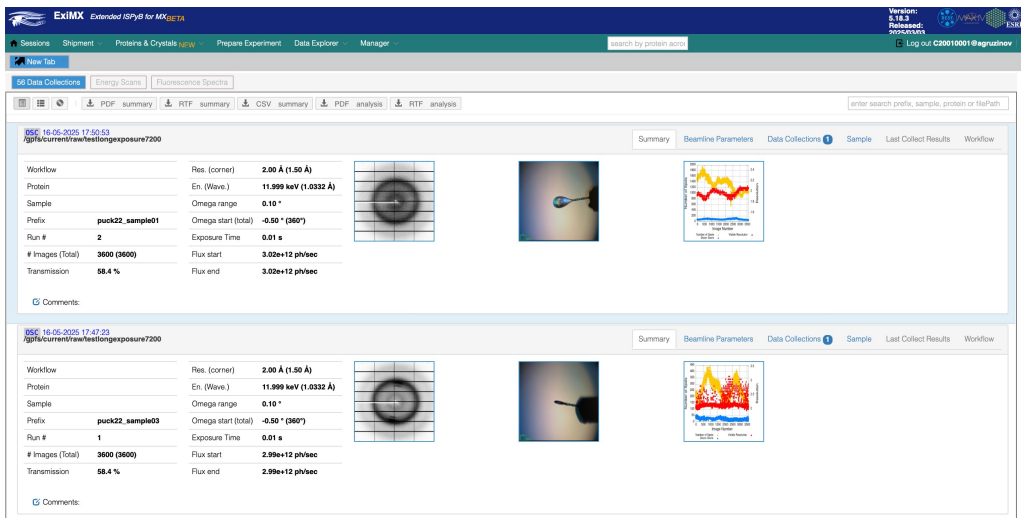
- Written in C++, Eiger Monitoring interface (10Hz)
- Automatically fetches images from the detector in real time
- Adds beam centre, energy, and detector distance for correct display
- Launches ADXV in socket mode for continuous updates during experiments.
- Based on EIGERStreamReceiver (Sascha Grimm)

User Benefits

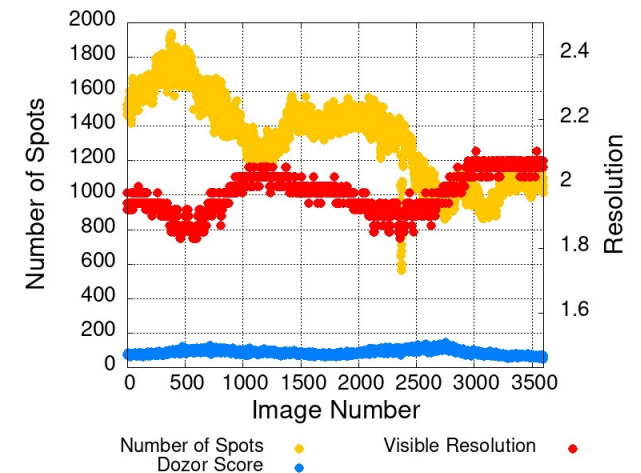
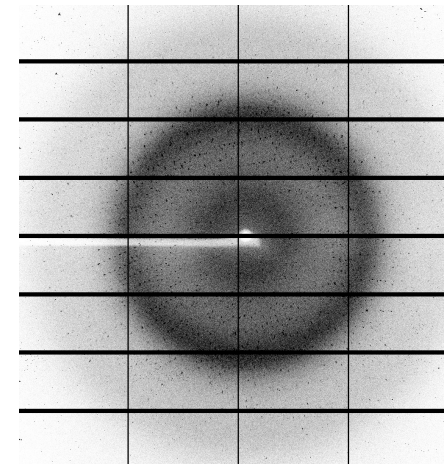
- Enables users and staff to assess data quality immediately.
- Familiar ADXV interface ensures a gentle learning curve for new users.

<https://github.com/agruzinov/yamone>

Synchronizing Metadata with EXI/ISPyB



- The old interface to LIMS is in use
- Snapshot generator – C++ HDF5 parser for fast saving – starts simultaneously with data processing
- Set up N frames to sum and contrast setting
- Image indicators – SBATCH script with parallel DOZOR run for each file frame
- Gnuplot for rendering



EDNA - diffraction characterization & data collection strategy calculation

Collaboration with G. Bourenkov (EMBL Hamburg) and Olof Svensson (ESRF)

Current status

- **Characterization Implementation Progress:**
 - Installed and integrated online data analysis framework EDNA and dependencies on Maxwell
 - Prototype characterization and strategy implementation at P11
 - Characterization is usable without dose estimation
- **Flux Estimator:**
 - Developed a dedicated flux estimation component tailored to the needs of the P11 beamline.
 - Estimation of photon flux based on fixed beam parameters.

Upcoming tasks

- **Migration to EDNA2:**
 - Transition data analysis processes to the latest version of EDNA2.
 - Ensure compatibility with existing workflows and software dependencies.

Developer Experience

- Great updates in documentation
- Constant feedback and code camps
- Extensive PR reviews and recommendations
- EMBL Hamburg experience – thanks for sharing!
- Martin's involvement in the Murko adaptation and advice

Problems Yet to Solve



- Migration to Python ≥ 3.12 + Qt updates
- Raster scans
- X-ray centring
- Murko centring optimization at P11
- Dev version follow-up
- XML/YAML upgrades
- Change of LIMS
- ENDA-based data processing
- Add XRF/Chooch?

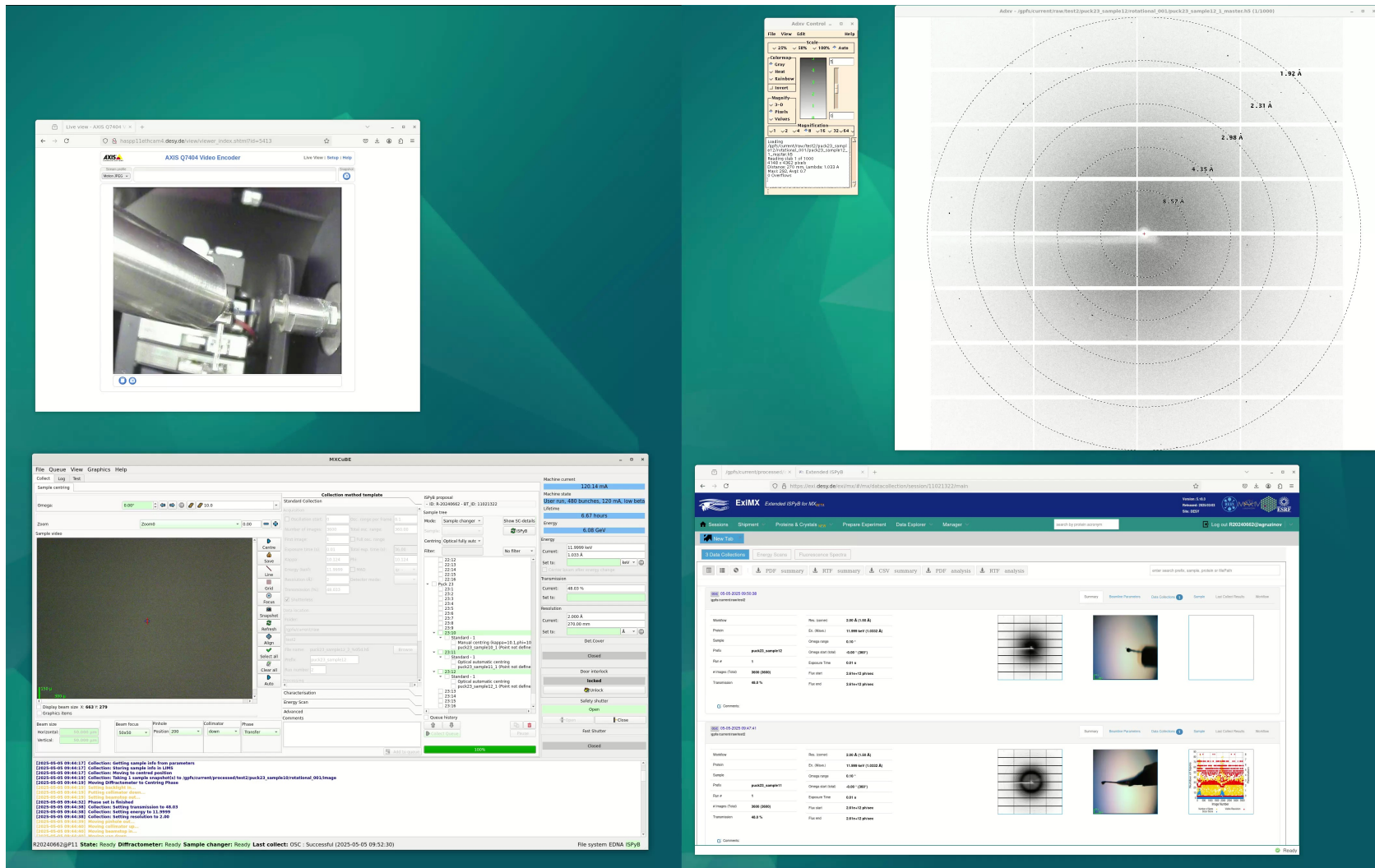


Summary

- **LIMS was installed and went into use**
- **Remote Data Collection re-established with MXCuBE**
- **Hardware Upgrade:**
 - Dedicated nodes for online data processing for faster feedback and auto-centring
- **Data processing and analysis have been completely moved to the external cluster**
- **MXCuBE Integration:**
 - Unified experiment control GUI with the familiar interface and ISPyB integration
- **Automation:**
 - Moving towards partial unattended data collection out of hours

Outlook

- **Short-term focus:** Achieve robust, reliable operations for P11 users and follow up on open PR reviews
- **Medium-term goals:**
 - Gradually reopen the codebase to upstream updates and new features
 - Re-align with mainline MXCuBE development branches
- **Long-term perspective:**
 - Explore the development of a digital twin or local test environment
 - Collaborate with the MXCuBE community on better testing strategies
 - Improve agility for future upgrades and feature adoption
- **Community feedback**
 - Give and receive feedback and solutions on the common problems during and off beamtimes



Acknowledgments

P11 Team

Johanna Hakanpää
Guillaume Pompidor
Spyros Chatziefthymiou
Alexander Grebentsov
Olga Merkulova (FS-SC)
Katharina Majd (User office)
Jialing Song
Sreenivasula Naidu Potturu
Veronica Delsoglio
Helena Taberman
Clemente Borges
Jan Meyer
Thorben Melson



FS-PETRA-D

FS-EC

Marcus Fleck
André Rothkirch,
Johannes Blume
Linus Pithan

IT

Frank Schluenzen
Jürgen Hannappel
Martin Gasthuber

FS-SC

Thomas White
Sam Flewett
Tim Schoof

FS-CFEL

Philipp Middendorf

ESRF

Bixente Bakaikoa
Marcus Oscarsson
Antonia Beteva

Soleil

Martin Savko



MXCuBE collaboration



Kooperation für
Anwendung und Innovation
der HAW Hamburg und DESY

P11 Users



Hamburgische
Investitions- und
Förderbank

HELMHOLTZ



EMBL Hamburg
Gleb Bourenkov
Marina Nikolova
Ivars Karpics

